1 $f = 5c - 8$

Make c the subject of the formula.

$$f = 5c - 8$$
$$+8 \qquad +8$$

$$\frac{f+8}{5} = \frac{5c}{5}$$

$$c = \frac{f+8}{5}$$

$c = \frac{f+8}{5}$

(Total for question 1 is 2 marks)

2 $u = 4t - 21$

Make t the subject of the formula.

$$u = 4t - 21$$
$$+21 \qquad +21$$

$$\frac{u+21}{4} = \frac{4t}{4}$$

$$t = \frac{u+21}{4}$$

$t = \frac{u+21}{4}$

(Total for question 2 is 2 marks)

3 $x = 3y - 2$

Make y the subject of the formula.

$$x = 3y - 2$$
$$+2 \qquad +2$$

$$\frac{x+2}{3} = \frac{3y}{3}$$

$$y = \frac{x+2}{3}$$

$y = \frac{x+2}{3}$

(Total for question 3 is 2 marks)

4 $m = 5n + 2p$

Make p the subject of the formula.

$$m = 5n + 2p$$
$$-5n \quad -5n$$
$$\frac{m - 5n}{2} = \frac{2p}{2}$$
$$p = \frac{m - 5n}{2}$$

$p = \frac{m - 5n}{2}$

(Total for question 4 is 2 marks)

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5 $a = 3c - 2$

Make c the subject of the formula.

$$a = 3c - 2$$
$$+2 \quad +2$$
$$\frac{a + 2}{3} = \frac{3c}{3}$$
$$c = \frac{a + 2}{3}$$

$c = \frac{a + 2}{3}$

(Total for question 5 is 2 marks)

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6 $P = 3a + 3b$

Make a the subject of the formula.

$$p = 3a + 3b$$
$$-3b \quad -3b$$
$$\frac{p - 3b}{3} = \frac{3a}{3}$$
$$a = \frac{p - 3b}{3}$$

$a = \frac{p - 3b}{3}$

(Total for question 6 is 2 marks)

7 Make n the subject of $m = n^2 + 3$

$$m = n^2 + 3$$
$$-3 \qquad -3$$
$$m - 3 = n^2$$
$$n = \pm\sqrt{m-3}$$

accept $n = \sqrt{m-3}$

$n = \pm\sqrt{m-3}$

(Total for question 7 is 2 marks)

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8 Make a the subject of $v = u + at$

$$v = u + at$$
$$-u \qquad -u$$
$$\frac{v-u}{t} = \frac{at}{t}$$
$$a = \frac{v-u}{t}$$

$a = \frac{v-u}{t}$

(Total for question 8 is 2 marks)

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9 Make a the subject of $v^2 = u^2 + 2as$

$$v^2 = u^2 + 2as$$
$$-u^2 \qquad -u^2$$
$$\frac{v^2 - u^2}{2s} = \frac{2as}{2s}$$
$$a = \frac{v^2 - u^2}{2s}$$

$a = \frac{v^2 - u^2}{2s}$

(Total for question 9 is 2 marks)

10 Make b the subject of $a = \sqrt{\frac{b+2}{5}}$

$$a^2 = \left(\sqrt{\frac{b+2}{5}}\right)^2$$

$$5 \times a^2 = \frac{b+2}{5} \times 5$$

$$5a^2 = b + 2$$

$$-2 \qquad -2$$

$$5a^2 - 2 = b$$

$b = 5a^2 - 2$

(Total for question 10 is 3 marks)

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11 Make b the subject of $A = 3b + 9$

$$A = 3b + 9$$

$$-9 \qquad -9$$

$$\frac{A-9}{3} = \frac{3b}{3}$$

$$b = \frac{A-9}{3}$$

$b = \frac{A-9}{3}$

(Total for question 11 is 2 marks)

12 Make x the subject of $y = 3x - 2$

$$y = 3x - 2$$
$$+2 \qquad +2$$
$$\frac{y+2}{3} = \frac{3x}{3}$$
$$x = \frac{y+2}{3}$$

$x = \frac{y+2}{3}$

(Total for question 12 is 2 marks)

13 Make x the subject of $y = \frac{1}{2}x + 6$

$$y = \frac{1}{2}x + 6$$
$$-6 \qquad -6$$
$$y - 6 = \frac{1}{2}x$$
$$\times 2 \qquad \times 2$$
$$2(y-6) = x$$

OR $x = 2y - 12$

$x = 2(y-6)$

(Total for question 13 is 2 marks)

14 Make x the subject of $y = \frac{2}{5}x - 12$

$$y = \frac{2}{5}x - 12$$
$$+12 \qquad +12$$
$$y + 12 = \frac{2}{5}x$$
$$\times 5 \qquad \times 5$$
$$\frac{5(y+12)}{2} = \frac{2x}{2}$$
$$x = \frac{5(y+12)}{2}$$

$x = \frac{5(y+12)}{2}$

(Total for question 14 is 3 marks)

15 Make x the subject of $5x + 6y + 12 = 0$

$-6y \quad -6y$

$5x + 12 = -6y$

$-12 \quad -12$

$5x = -6y - 12$

$x = \frac{-6y - 12}{5}$

$x = \frac{-6y - 12}{5}$

(Total for question 15 is 2 marks)

16 Make x the subject of $y = x^3 - 5$

$+5 \quad +5$

$y + 5 = x^3$

$x = \sqrt[3]{y + 5}$

$x = \sqrt[3]{y + 5}$

(Total for question 16 is 2 marks)

17 Make x the subject of $y = \frac{2x + 3}{4}$

$\times 4 \quad \times 4$

$4y = 2x + 3$

$-3 \quad -3$

$\frac{4y - 3}{2} = \frac{2x}{2}$

$x = \frac{4y - 3}{2}$

$x = \frac{4y - 3}{2}$

(Total for question 17 is 3 marks)

18 Make a the subject of $x = 3(a + 9)$

$x = 3a + 27$

$-27 \qquad -27$

$\frac{x - 27}{3} = \frac{3a}{3}$

$a = \frac{x - 27}{3}$

OR $a = \frac{x}{3} - 9$

$a = \frac{x - 27}{3}$

(Total for question 18 is 2 marks)

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19 $a = \frac{3 + c}{b}$

Make b the subject of the formula.

$ab = 3 + c$

$b = \frac{3 + c}{a}$

$b = \frac{3 + c}{a}$

(Total for question 19 is 2 marks)

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20 $d = \sqrt{\frac{3h}{2}}$

Make h the subject of the formula.

$d^2 = \frac{3h}{2}$

$\frac{2d^2}{3} = \frac{3h}{3}$

$h = \frac{2d^2}{3}$

$h = \frac{2d^2}{3}$

(Total for question 20 is 3 marks)